



- DESX was proposed by R. Rivest to protect DES against exhaustive key search. It uses one 64-bit secret key W and a 56-bit DES key K . DESX operates as follows:
 $C = W \oplus E_K(P \oplus W)$. Show how the decryption is done
- We consider the block cipher mode CBC, where an n block message $M_1M_2\dots M_n$ is encrypted to an $n+1$ block message $C_0C_1C_2\dots C_n$, where: $C_0 = IV$ and
$$C_k = E_K(M_k \oplus C_{k-1}), k = 1, 2, \dots, n.$$
 - How is decryption affected if the C_0 is removed from the ciphertext?
 - How is decryption affected if C_n is removed from the ciphertext?
- We already have the encryption of a message as above. We want to add one specified plaintext block M_0 in the beginning of the message, so that the extended message is $M_0M_1M_2\dots M_n$. Show that a legitimate user (someone who knows the key K) can achieve this by just adding one ciphertext block C_{-1} in the beginning of the ciphertext, to get encryption $C_{-1}C_0C_1C_2\dots C_n$.
- Let (E, D) be a cipher. Write the correct decryption in the schemes

- $E'((k_1, k_2), m) = E(k_1, m) \oplus k_2 \oplus k_1.$
 - $E'((k_1, k_2, k_3), m) = E(k_3, E(k_2, E(k_1, m))).$

 formula for encryption
- A sequence of plaintext blocks $M_1, \dots, M_8$ is encrypted using DES into a sequence of ciphertext blocks. Where an IV is used, it is numbered C_0 . A transmission error occurs and **one bit in ciphertext block C_3 changes its value**. As a consequence, the receiver obtains after decryption a corrupted plaintext block sequence $M'_1, \dots, M'_8$. For the discussed modes of operation (ECB, CBC, CFB, OFB, CTR), how many bits do you expect to be wrong in each block M'_i ?
- Consider a block cipher named *simpleCipher*, which operates as per Algorithm-1

Algorithm-1 *simpleCipher* (*key*, *m*)**Require:** key and m are 4 bits long $z = \text{key XOR } m$ **return** *lookuptable*(z)

The *lookuptable* function (in the algorithm) is illustrated in the following table (Table1)

Table 1- *lookuptable* function

Input	Output	Input	Output	Input	Output	Input	Output
0000	0011	0100	0111	1000	1100	1100	1000
0001	0100	0101	1001	1001	1110	1101	0001
0010	0101	0110	1101	1010	1011	1110	0010
0011	0110	0111	1111	1011	1010	1111	0000

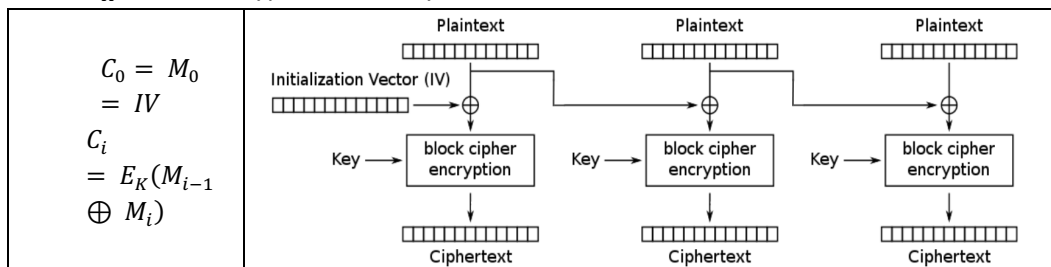
The cipher is applied using the counter mode of operation (CTR) to encrypt a plaintext $P=10101100$ with key $k = 1011$. The counter ctr_i is constructed as the concatenation of a *nonce* (2 bits) and the running counter i (over 2 bits) that starts at 1; let the nonce be 00 . Calculate the values of the ciphertext; show the different steps



7. Amani and Batoul invented a simple cipher to protect their messages from being intercepted by Eva. They called it the ABC cipher. The cipher takes a block of 4 bits and a key of 4 bits K and encrypts it as follows: Let $M = b_1b_2b_3b_4$ a plaintext block, the encryption steps are as follows:

1. Calculate $s = (K)_{10} \bmod 4$, where $(K)_{10}$ is the decimal value of K
2. Apply a **left** circular shift of s bits to M to obtain M'
3. $E_k(M) = M' \oplus k$

- a) Explain how the decryption algorithm works.
 - b) Amani and Batoul decided to use (7) their lucky number as an encryption key. Amani wants to send the message “Hi” to Batoul. She encoded it in ASCII to obtain the following binary sequence: 0100 1000 0110 1001. Calculate the corresponding ciphertext.
 - c) Explain how Batoul will decrypt the letter ‘H’ of the message. Show your calculation.
 - d) In order to avoid the problems related to the ECB mode, Amani and Batoul decided to use the CBC mode (recall that the CBC works as follows: $C_i = E_k(M_i \oplus C_{i-1}, C_0 = IV)$). They use the value 0100 as IV. Batoul wants to reply with a “Hi” message to Amani. Explain how the encryption process works (show your calculation)
8. Consider the following block cipher mode of operation: M_i is the i^{th} plaintext block. C_i is the i^{th} ciphertext block. E_K is AES encryption with key K .



- a) Write the decryption equation for the operation mode
- b) Is the encryption parallelizable? Justify your answer
- c) Is the decryption parallelizable? Justify your answer
- d) Ali encrypted a message M that consists of eight blocks: $M_1M_2M_3 \dots M_8$. He obtained the ciphertext $C = C_0C_1C_2C_3 \dots C_8$. Suppose that he decided to re-encrypt the message after modifying one bit of first block M_1 . How the ciphertext blocks will be affected? In other words, explain which ciphertext blocks will be modified and which remained the same.
- e) Now suppose that Ali sent the ciphertext $C = C_0C_1C_2C_3 \dots C_8$ to Bilal. During transmission, one bit of C_2 was affected. What are the message blocks that will be affected after decryption. Explain.